Table 52. I/O AC characteristics (continued)

Speed	Symbol	Parameter[1][2]	Conditions	Min	Max	Unit
01	Fmax	Maximum frequency	C=50 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	10.00	MHz
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	2.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	15.00	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	2.50	
	Tr/Tf	Output rise and fall time[3]	C=50 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	30.00	ns
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	60.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	15.00	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	30.00	
10	Fmax	Maximum frequency	C=50 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	30.00	MHz
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	15.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	60.00[4]	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	30.00	
	Tr/Tf	Output rise and fall time[3]	C=50 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	11.00	ns
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	22.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	4.00	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	8.00	
11	Fmax	Maximum frequency	C=30 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	60.00[4]	MHz
			C=30 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	30.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	80.00[4]	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	40.00	
	Tr/Tf	Output rise and fall time[3]	C=30 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	5.50	ns
			C=30 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	11.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	2.50	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	5.00	

1. The I/O speed is configured using the OSPEEDRy[1:0] bits. The Fm+ mode is configured in the SYSCFG_CFGR1 register. Refer to the RM0516 reference manual for a description of GPIO Port configuration register.
2. Specified by design. Not tested in production.
3. The fall time is defined between 70% and 30% of the output waveform, according to I^2C specification.
4. This value represents the I/O capability but the maximum system frequency is limited to 48 MHz.